

# FAULT CODE 622 - Exhaust Gas Temperature Deviation Low for Cylinder 3 - Data Valid But Below Normal Operating Range - Least Severe Level

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## Troubleshooting Summary

| STEPS   |                                                                                        | SPECIFICATIONS                                                                                                                                                                 |
|---------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STEP 1. | Check the exhaust gas temperature sensor circuit cylinder 3 accuracy.                  |                                                                                                                                                                                |
|         | <b>STEP 1A.</b> Verify the exhaust gas temperature sensor circuit cylinder 3 accuracy. | Temperature measurement from the infrared thermometer within 15 percent of the exhaust gas temperature sensor circuit cylinder 1 reading with INSITE™ electronic service tool? |
| STEP 2. | Clear the fault code.                                                                  |                                                                                                                                                                                |
|         | <b>STEP 2A.</b> Disable the fault code.                                                | Fault Code 622 inactive?                                                                                                                                                       |
|         | <b>STEP 2B.</b> Clear any inactive fault codes.                                        | All fault codes cleared?                                                                                                                                                       |

**STEP 1. Check the exhaust gas temperature sensor circuit cylinder 3 accuracy.**

**STEP 1A. Verify the exhaust gas temperature sensor circuit cylinder 3 accuracy.**

**Conditions :**

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

| Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Specification/Repair                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Next Step                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <p>Verify the exhaust gas temperature sensor circuit cylinder 3 accuracy with an infrared thermometer.</p> <ul style="list-style-type: none"> <li>• Start the engine.</li> <li>• Connect INSITE™ electronic service tool.</li> <li>• Use an infrared thermometer to measure and record the surface temperature at the exhaust port side of the cylinder head.</li> <li>• Compare the infrared thermometer temperature measurement with the exhaust gas temperature sensor circuit cylinder 3 reading on the INSITE™ electronic service tool monitor screen.</li> </ul> | <p>Temperature measurement from the infrared thermometer within 15 percent of the exhaust gas temperature sensor circuit cylinder 1 reading with INSITE™ electronic service tool?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Possible cylinder or injector damage <b>must</b> be investigated.</p>                                                                                                                                                                                                                                        | <p><b>Engine Performance Troubleshooting Tree</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Temperature measurement from the infrared thermometer within 15 percent of the exhaust gas temperature sensor circuit cylinder 1 reading with INSITE™ electronic service tool?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>A faulty exhaust gas temperature sensor circuit cylinder 3 has been detected.</p> <p>Replace the exhaust gas temperature sensor circuit cylinder 3.</p> <ul style="list-style-type: none"> <li>• Refer to Procedure 019-013 in Section 19.<br/>(/qs3/pubsys2/xml/en/procedures/81/81-019-013.html)</li> </ul> | <p><b>2A</b></p>                                      |

## STEP 2. Clear the fault codes.

### STEP 2A. Disable the fault code.

**Conditions :**

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

| Action                                                                                                                                              | Specification/Repair                                                                                                                                                                                         | Next Step |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Disable the fault code. <ul style="list-style-type: none"><li>• Use INSITE™ electronic service tool to verify the fault code is inactive.</li></ul> | Fault Code 622 inactive?<br><b>YES</b>                                                                                                                                                                       | <b>2B</b> |
|                                                                                                                                                     | Fault Code 622 inactive?<br><b>NO</b><br><br><b>Repair :</b><br><br>Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again. | <b>1A</b> |

### STEP 2B. Clear the inactive fault codes.

**Conditions :**

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

| Action                                                                                                                                                   | Specification/Repair                   | Next Step                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------------------------------|
| Clear the inactive fault codes. <ul style="list-style-type: none"><li>• Use INSITE™ electronic service tool to clear any inactive fault codes.</li></ul> | All fault codes cleared?<br><b>YES</b> | <b>Repair complete</b>                   |
|                                                                                                                                                          | All fault codes cleared?<br><b>NO</b>  | <b>Appropriate troubleshooting steps</b> |

